

The Risk Protection Value of “Moral Hazard”

Angie Acquatella
TSE

Victoria Marone
Yale & NBER

April 2026

Terminology

- Classic empirical result in health economics :

Higher coverage health insurance causes greater consumption of healthcare

Terminology

- Classic empirical result in health economics :

Higher coverage health insurance causes greater consumption of healthcare
moral hazard utilization

Terminology

- Classic empirical result in health economics :

Higher coverage health insurance causes greater consumption of healthcare
moral hazard utilization

“From a normative point of view, moral hazard [utilization] can be argued to cause a negative externality to the extent that it causes the insurer ... to increase premiums. Thus, moral hazard [utilization] should be avoided.”

Zweifel and Manning, 2000 (*Handbook of Health Economics*)

Terminology

- Classic empirical result in health economics :

Higher coverage health insurance causes greater consumption of healthcare
moral hazard utilization

Terminology

- Classic empirical result in health economics :

Higher coverage health insurance causes greater consumption of healthcare
moral hazard utilization

→ as distinct from *moral hazard* : asymmetric info between contracting parties
with non-aligned incentives

Terminology

- Classic empirical result in health economics :

Higher coverage health insurance causes greater consumption of healthcare
moral hazard utilization

→ as distinct from *moral hazard* : asymmetric info between contracting parties
with non-aligned incentives

⇒ The risk protection value of “*moral hazard*”

=

The risk protection value of *moral hazard utilization*

=

The risk protection value of *the ability to change behavior in response to insurance*

Motivation

- ★ Classic insight from optimal insurance literature : in a **second-best** world,
value of risk protection must be weighed against cost of distorted incentives
- But insurance may change behavior even without distorting incentives

Motivation

- ★ Classic insight from optimal insurance literature : in a **second-best** world,
value of **risk protection** must be weighed against cost of **distorted incentives**
- But insurance may change behavior even without distorting incentives
- Why are we distorting incentives? → Can't contract on **health state**

Motivation

- ★ Classic insight from optimal insurance literature : in a **second-best** world, value of **risk protection** must be weighed against cost of **distorted incentives**
- But insurance may change behavior even without distorting incentives
- Why are we distorting incentives? → Can't contract on **health state**
 - ▶ in **first best** world : full and symmetric information (complete markets)
 - **state-contingent contracts** (Arrow-Debreu securities)
 - can move resources (income) without distorting incentives (i.e., $MRS=MRT$)→ and moving resources might change behavior (healthcare utilization)
 - ▶ in **second best** : asymmetric info (incomplete markets) ↔ **moral hazard**
 - **indemnity contracts** (“makes you whole” in event of covered perils)
 - can only distort prices (incentives) (i.e., $MRS \neq MRT$)→ which might also change behavior ↔ **moral hazard utilization**

Research question

How should we think about, from a normative perspective,
the changes in behavior that result from insurance contracts
in a second-best world?

Research questions and answers

(i) Can moral hazard utilization be socially valuable in health insurance?
↳ in sense that planner would not prohibit, if it could

(ii) Under what conditions?

(iii) Quantitatively, to what extent?

Research questions and answers

(i) Can moral hazard utilization be socially valuable in health insurance?

↳ in sense that planner would not prohibit, if it could

▶ Yes, because it facilitates risk protection

→ intuition : actuarially fair price reduction shifts real income across states

(ii) Under what conditions?

(iii) Quantitatively, to what extent?

Research questions and answers

(i) Can moral hazard utilization be socially valuable in health insurance?

↳ in sense that planner would not prohibit, if it could

▶ Yes, because it facilitates risk protection

→ intuition : actuarially fair price reduction shifts real income across states

(ii) Under what conditions?

▶ Necessary : risk and non-zero income elasticity of demand for healthcare

▶ Hinges on : size of desired resource transfer towards sick states +
relative size of subst vs income effect from healthcare price decrease

(iii) Quantitatively, to what extent?

Research questions and answers

(i) Can moral hazard utilization be socially valuable in health insurance?

↳ in sense that planner would not prohibit, if it could

- ▶ Yes, because it facilitates risk protection

→ intuition : actuarially fair price reduction shifts real income across states

(ii) Under what conditions?

- ▶ Necessary : risk and non-zero income elasticity of demand for healthcare
- ▶ Hinges on : size of desired resource transfer towards sick states +
relative size of subst vs income effect from healthcare price decrease

(iii) Quantitatively, to what extent?

- ▶ Standard estimates imply large risk protection value of moral hazard utilization
→ prohibiting moral hazard utilization en masse *lowers* social welfare

Related literature

- ★ Closely related work on value of “income effect” (Marshall 1976; de Meza 1982; Nyman 1999/2001/2018)
 - ▶ here : formalizing this intuition → link with risk protection
- Foundational work on optimal insurance (Arrow 1965; Pauly 1968; Zeckhauser 1970)
 - ▶ classic trade-off : risk protection vs. cost of distorted incentives
 - ▶ we emphasize : **behavioral response can be a part of how risk protection is produced**

→ larger behavioral response $\neq$ larger cost of distorted incentives
- Work highlighting other reasons moral hazard utilization may be valuable (Baicker et al 2015 [internalities/info frictions]; Chernow Frick 2009 [market power]; Newhouse 2006 [externalities]; [Ericson Sydnor Jaspersen 2025](#) [liquidity constraints])
 - ▶ here : only channel is risk protection; risk is necessary
- Work quantifying welfare consequences of moral hazard in health insurance (e.g., Pauly 1969; Manning et al 1987; Feldman Dowd 1991; Finkelstein Hendren Luttmer 2019)
 - ▶ here : show (and quantify) how moral hazard utiliz. can enter *both* positively and negatively

1. Introduction

2. Model & empirical magnitudes

3. Discussion & conclusion

1. Introduction

2. Model & empirical magnitudes

3. Discussion & conclusion

Model outline

Math

- 1 Consumers
- 2 First-best benchmark
- 3 Feasible insurance contracts
- 4 Welfare

Pictures

- 1.a Ordinal preferences
- 1.b Risk preferences
- 2.a First-best allocations
- 2.b Change in behavior under first best
- 3.a Moral hazard utilization
- 3.b Decomposition of utilization
- 4.a Overall
- 4.b Decomposition of welfare

Model I: Consumers

- Consumer has state-dependent preferences represented by vNM utility $u(y, m; l)$

$y \in \mathbb{R}_+$: non-health consumption

$m \in \mathbb{R}_+$: healthcare consumption (aka utilization, or spending)

$l \in L$: health state

where $\forall l: u_y > 0$, u concave in (y, m) , bliss point in m , $\frac{u_m}{u_y}|_l > 0 \rightarrow$ so higher l = sicker

Model I: Consumers

- Consumer has state-dependent preferences represented by vNM utility $u(y, m; l)$

$y \in \mathbb{R}_+$: non-health consumption

$m \in \mathbb{R}_+$: healthcare consumption (aka utilization, or spending)

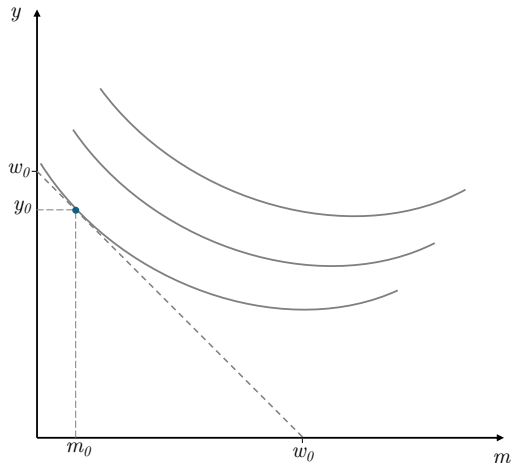
$l \in L$: health state

where $\forall l: u_y > 0$, u concave in (y, m) , bliss point in m , $\frac{u_m}{u_y}|_l > 0 \rightarrow$ so higher l = sicker

- Units of y and m set such that relative market price (MRT) is 1
- Consumer faces **risk** over her health state $l \sim F \rightarrow$ and thus, her tastes for (y, m)
- Consumer endowed with initial resources $w_0 \rightarrow$ same in all states
- **Information** : F commonly known, realization l private to consumer
 - ↳ non-verifiable health status is singular market failure $\rightarrow$ no selection, just moral hazard

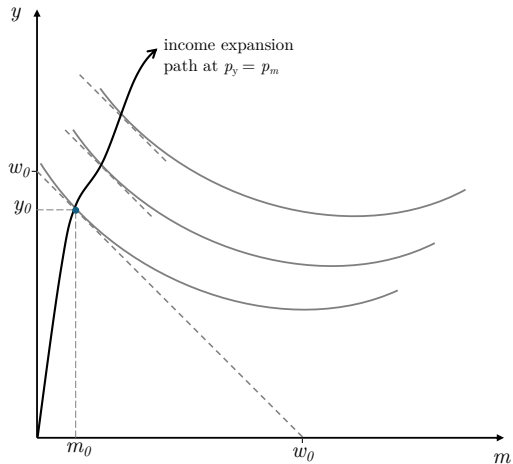
Pictures part 1a: Consumer's ordinal preferences over (y, m) given l

$$l = \hat{l}$$



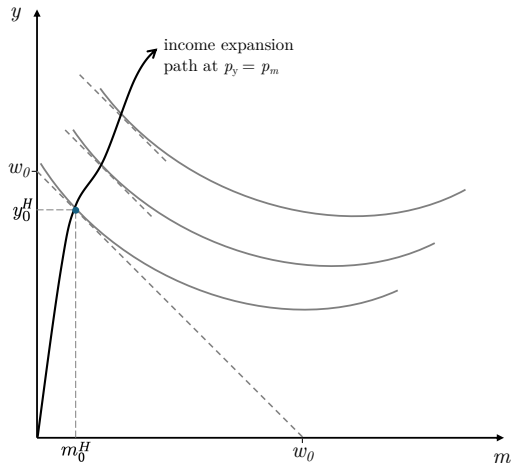
Pictures part 1a: Consumer's ordinal preferences over (y, m) given l

$$l = \hat{l}$$

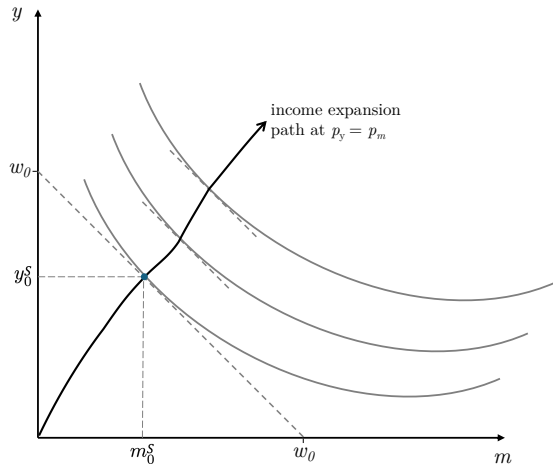


Pictures part 1a: Consumer's ordinal preferences over (y, m) given l

$l = \text{Healthy}$

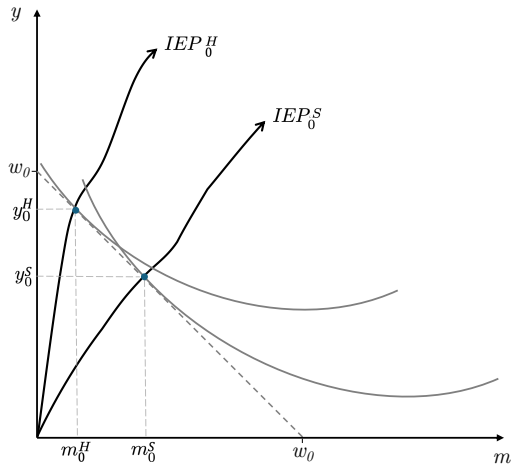


$l = \text{Sick}$



Pictures part 1a: Consumer's ordinal preferences over (y, m) given l

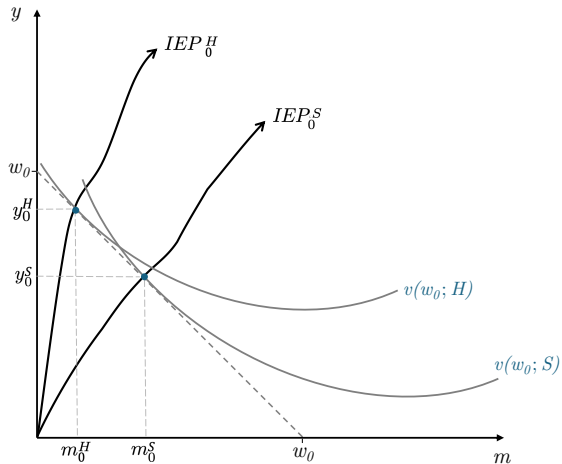
Ordinal preferences



$$\max_{y, m} u(y, m; l) \quad s.t. \quad y + m \leq w_0$$

Pictures part 1b: Consumer's risk preferences

Ordinal preferences

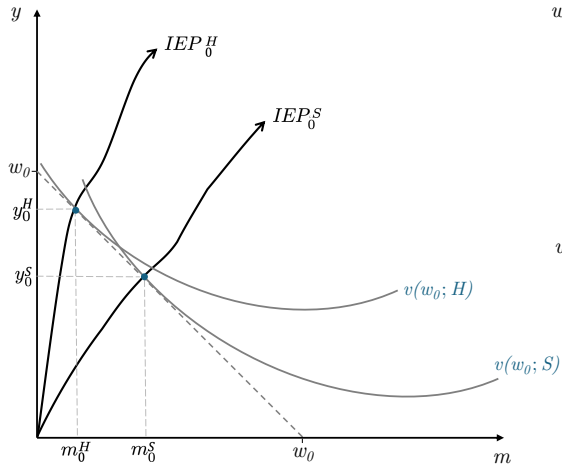


$$\max_{y, m} u(y, m; l) \quad s.t. \quad y + m \leq w_0$$
$$= v(w_0; l)$$

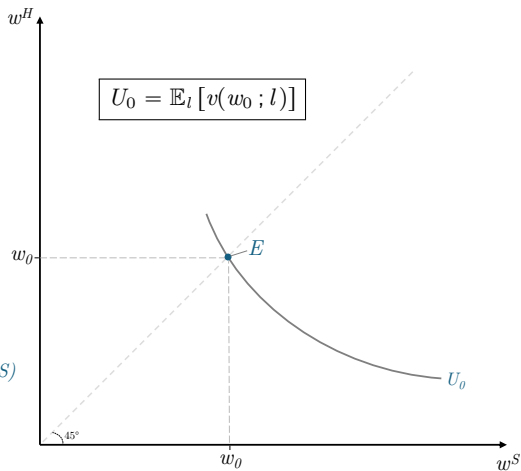
$$U_0 = \mathbb{E}_l [v(w_0; l)]$$

Pictures part 1b: Consumer's risk preferences

Ordinal preferences

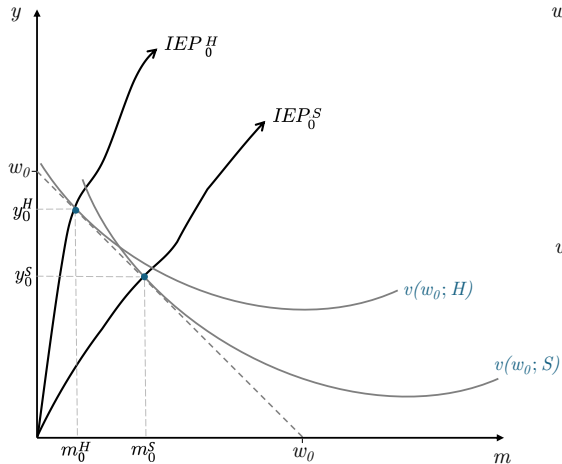


Risk preferences

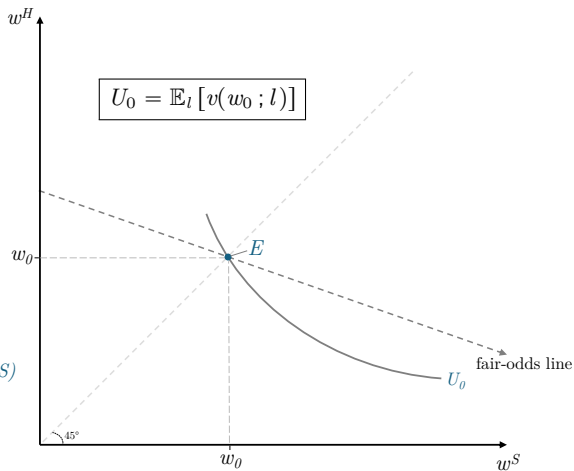


Pictures part 1b: Consumer's risk preferences

Ordinal preferences



Risk preferences



Model part 2: First-best benchmark

- Suppose health state were contractible $\rightarrow$ contract may be state-contingent
- In first-best world, consumer solves

$$\max_{w(l)} \mathbb{E}_l[v(w(l); l)] \quad \text{s.t.} \quad \mathbb{E}_l w(l) \leq w_0$$

$\hookrightarrow$ where $v(w; l)$ is indirect utility function given $p_y = p_m$

$$\Longleftrightarrow$$

$$\max_{y(l), m(l)} \mathbb{E}_l[u(y(l), m(l); l)] \quad \text{s.t.} \quad \mathbb{E}_l[y(l) + m(l)] \leq w_0$$

Model part 2: First-best benchmark

- Suppose health state were contractible $\rightarrow$ contract may be state-contingent
- In first-best world, consumer solves

$$\max_{w(l)} \mathbb{E}_l[v(w(l); l)] \quad \text{s.t.} \quad \mathbb{E}_l w(l) \leq w_0$$

$\hookrightarrow$ where $v(w; l)$ is indirect utility function given $p_y = p_m$

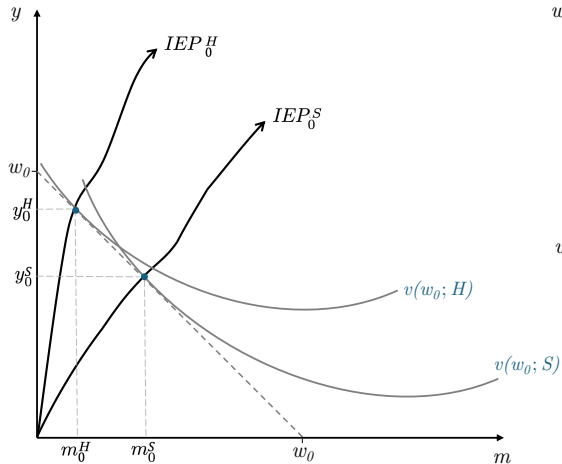
$$\Longleftrightarrow$$

$$\max_{y(l), m(l)} \mathbb{E}_l[u(y(l), m(l); l)] \quad \text{s.t.} \quad \mathbb{E}_l[y(l) + m(l)] \leq w_0$$

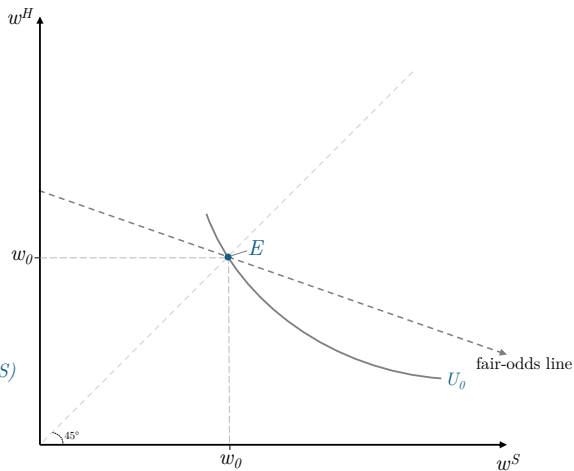
- Solution yields first-best allocations in each state : $w^{FB}(l) = y^{FB}(l) + m^{FB}(l)$
- $\Rightarrow$ Define $m^{FB}(l)$ to be the socially efficient level of healthcare utilization

Pictures part 2a: First-best allocations

Ordinal preferences

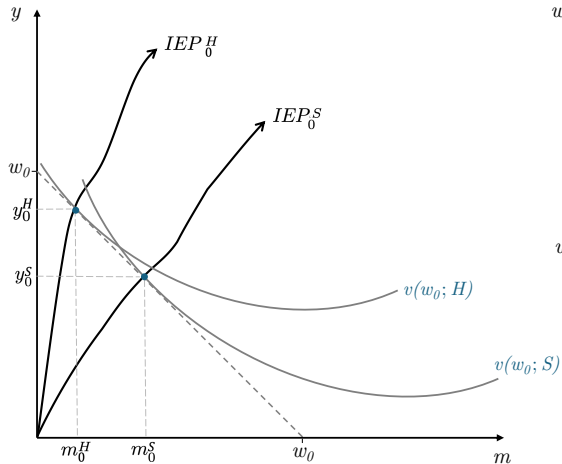


Risk preferences

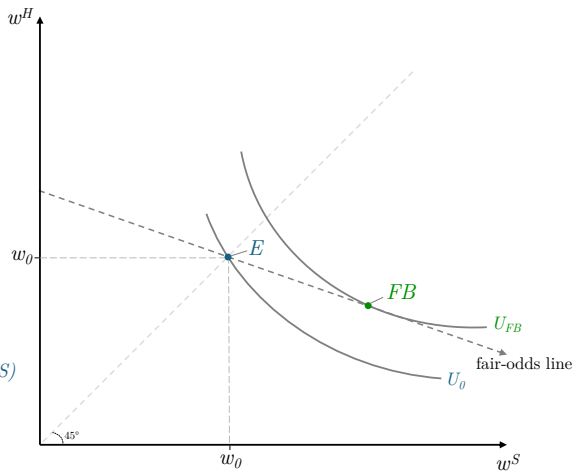


Pictures part 2a: First-best allocations

Ordinal preferences

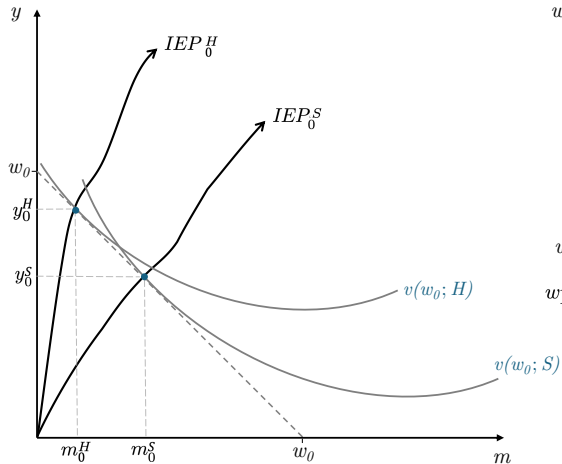


Risk preferences

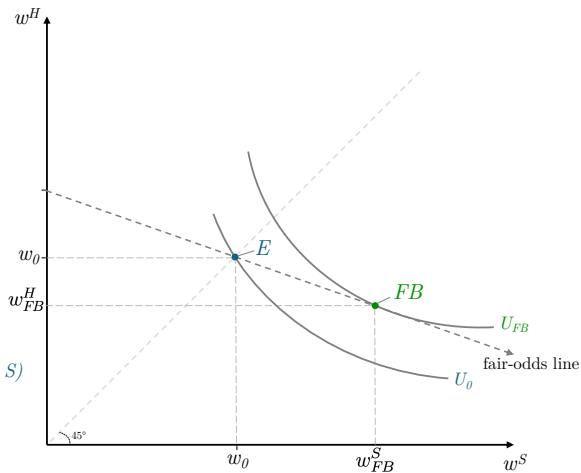


Pictures part 2a: First-best allocations

Ordinal preferences

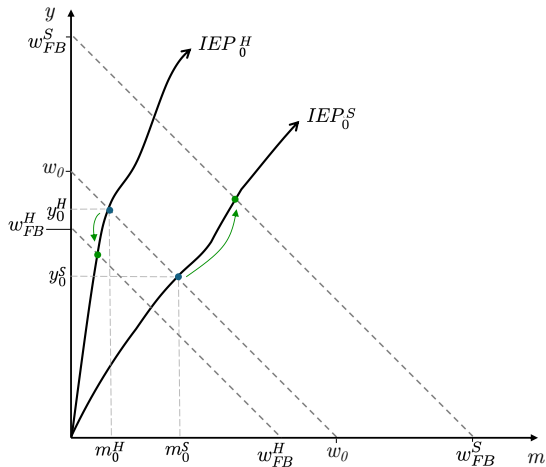


Risk preferences

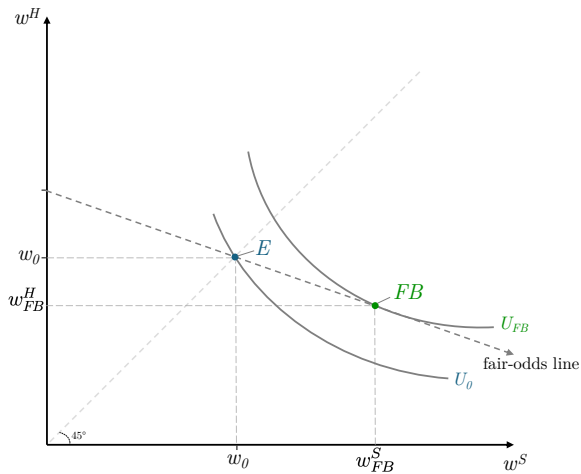


Pictures part 2a: First-best allocations

Ordinal preferences

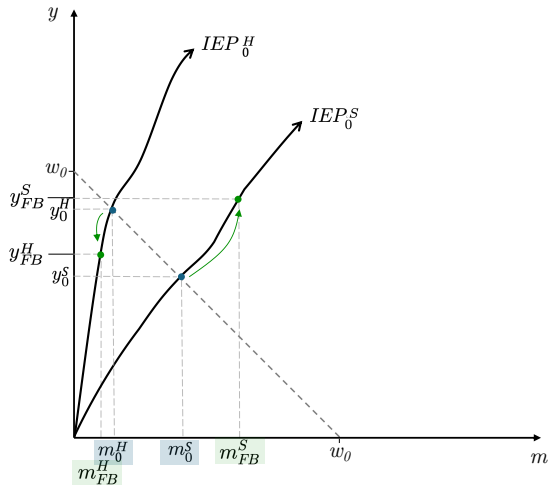


Risk preferences

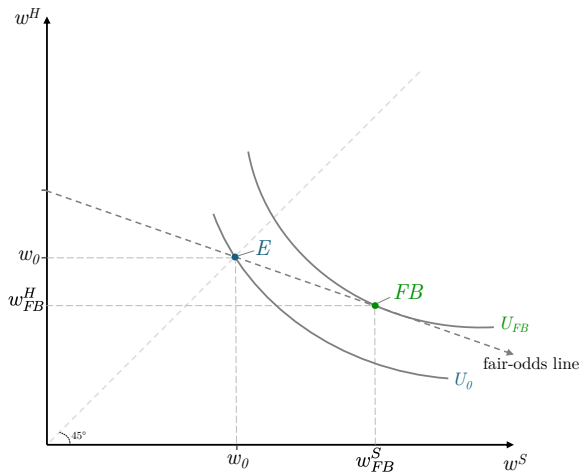


Pictures part 2a: First-best allocations

Ordinal preferences



Risk preferences



Model part 3: Feasible insurance contracts

Back in second-best world...

- Insurance contract x lowers consumer price of healthcare to $c(x) \in [0, 1]$
 - ↳ no insurance = null contract x_0 where $c(x_0) = 1$

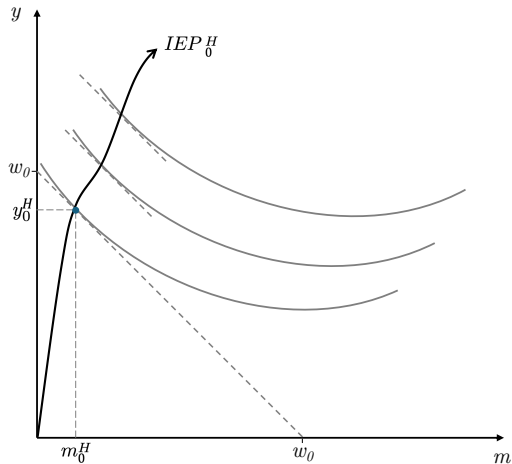
Model part 3: Feasible insurance contracts

Back in second-best world...

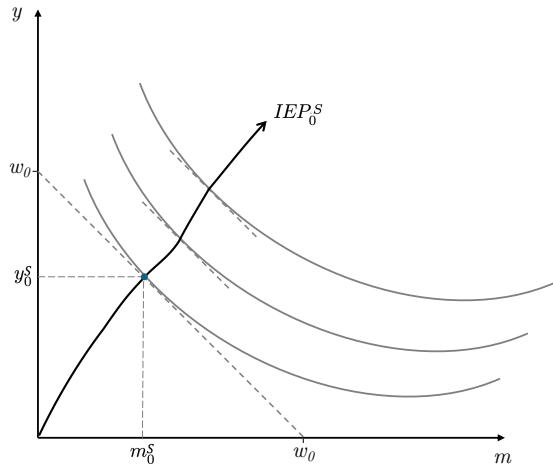
- Insurance contract x lowers consumer price of healthcare to $c(x) \in [0, 1]$
↳ no insurance = null contract x_0 where $c(x_0) = 1$
- Consumer solves : $\max_{y, m} u(y, m; l) \quad s.t. \quad y + m \cdot c(x) \leq w_0 - p$
 $\Rightarrow m^*(l, x, w_0, p) \quad \text{and} \quad y^*(l, x, w_0, p)$
- Actuarially fair premium : $\bar{p}(x) = \mathbb{E}_l[(1 - c(x)) \cdot m^*(l, x, w_0, \bar{p}(x))]$

Pictures part 3a: Feasible insurance contracts: Consider full insur.

$l = \text{Healthy}$

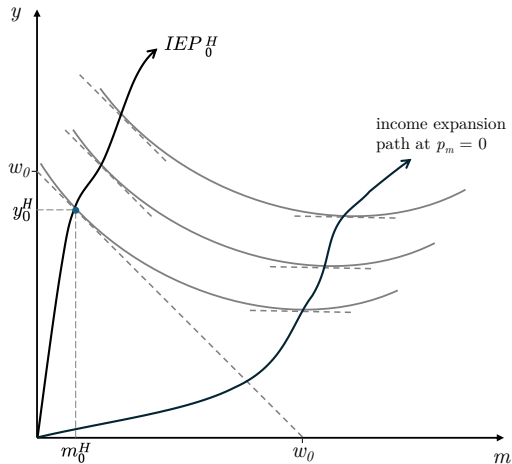


$l = \text{Sick}$

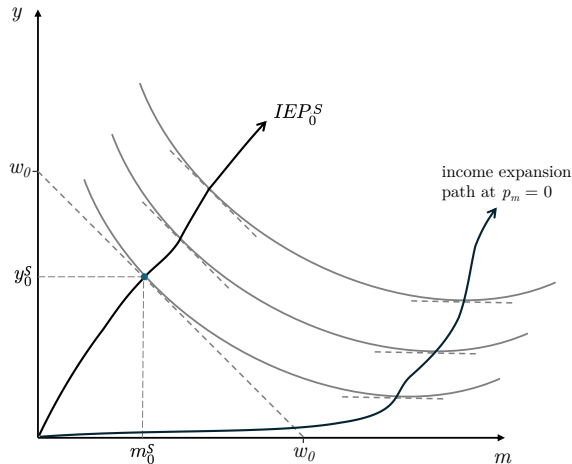


Pictures part 3a: Feasible insurance contracts: Consider full insur.

$l = \text{Healthy}$

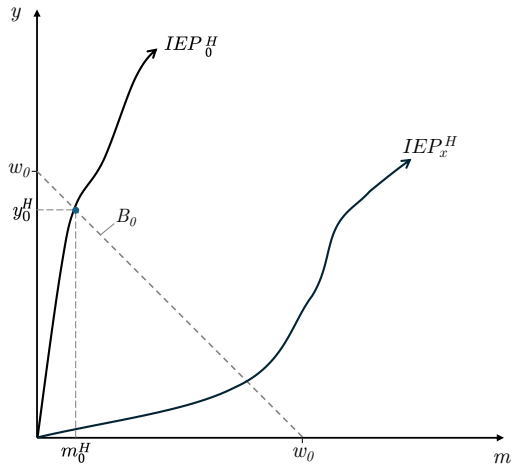


$l = \text{Sick}$

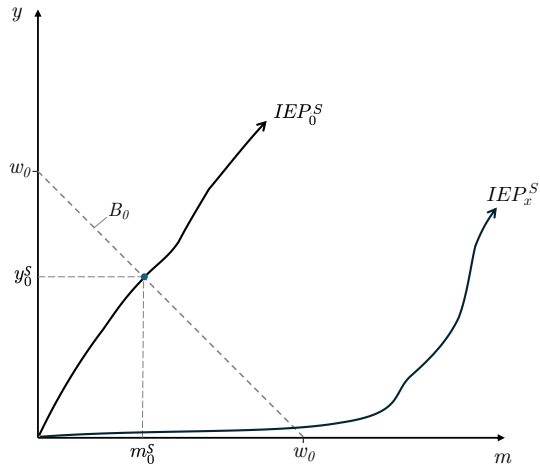


Pictures part 3a: Feasible insurance contracts: Consider full insur.

$l = \text{Healthy}$

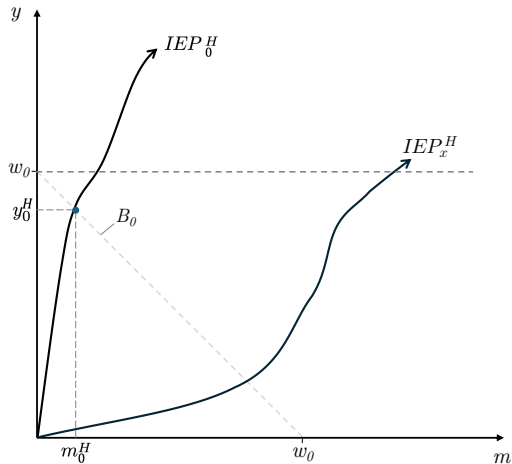


$l = \text{Sick}$

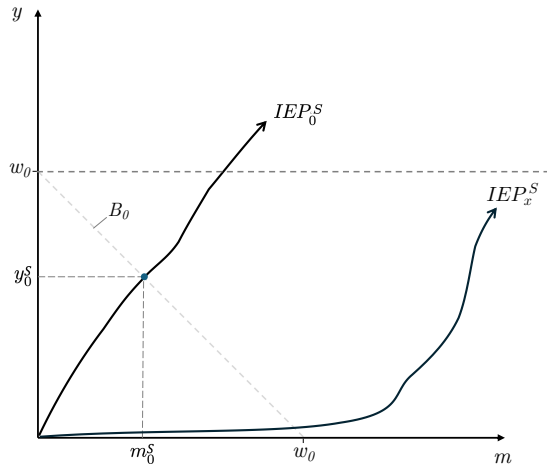


Pictures part 3a: Feasible insurance contracts: Consider full insur.

$l = \text{Healthy}$

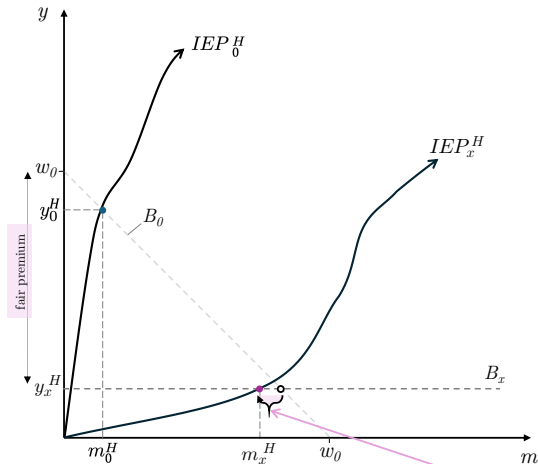


$l = \text{Sick}$

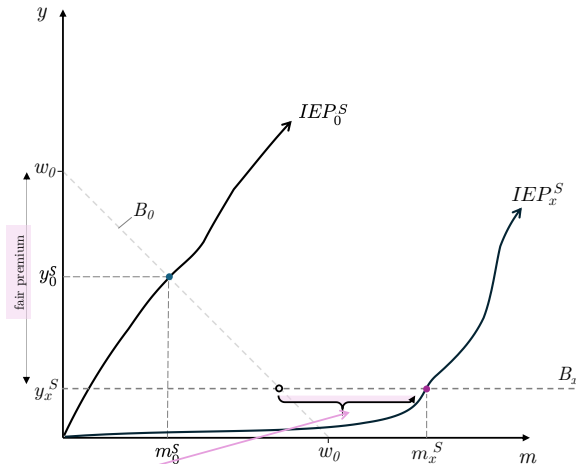


Pictures part 3a: Feasible insurance contracts: Consider full insur.

$l = \text{Healthy}$



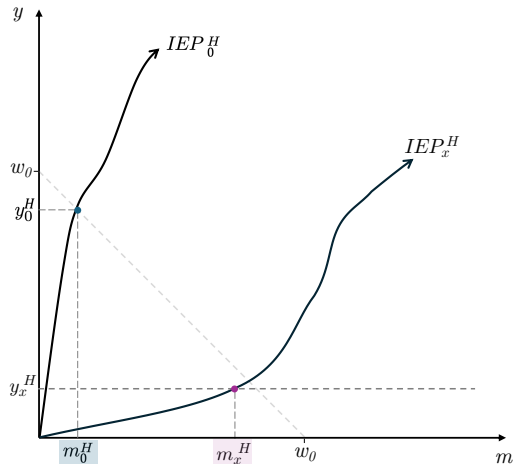
$l = \text{Sick}$



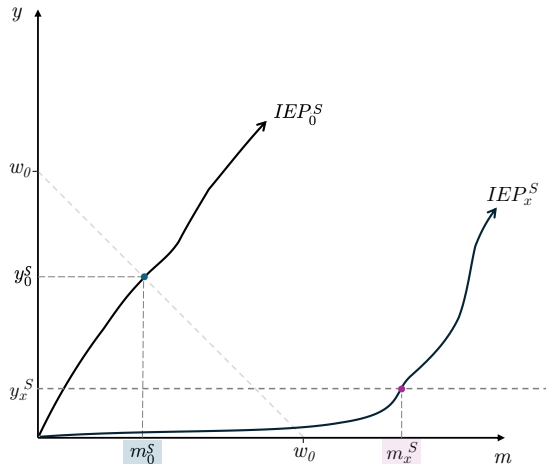
actuarially fair
deviations from
original budget set

Pictures part 3b: Decomposition of moral hazard utilization Slutsky

$l = \text{Healthy}$

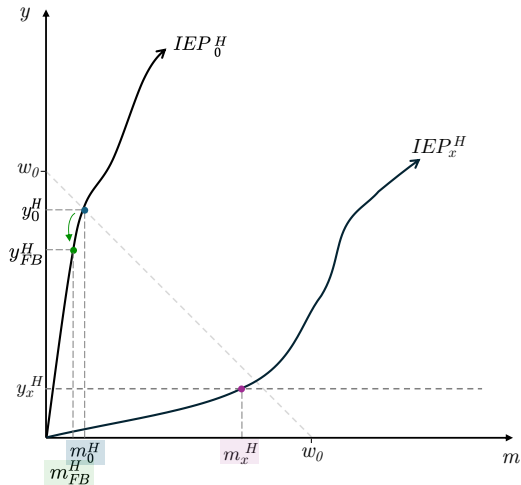


$l = \text{Sick}$

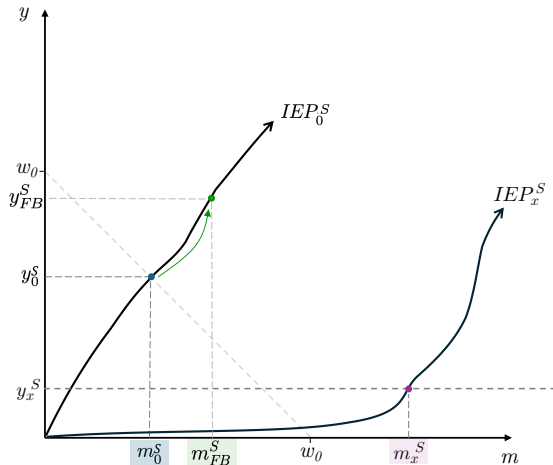


Pictures part 3b: Decomposition of moral hazard utilization Slutsky

$l = \text{Healthy}$

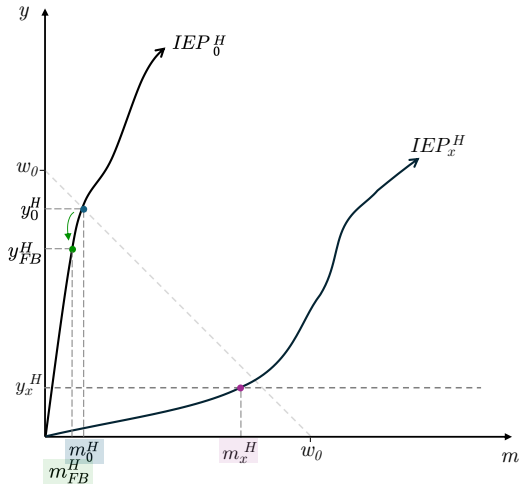


$l = \text{Sick}$

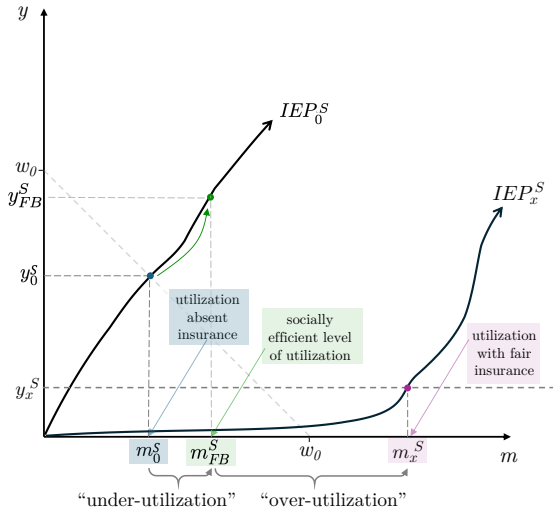


Pictures part 3b: Decomposition of moral hazard utilization Slutsky

$l = \text{Healthy}$



$l = \text{Sick}$



Model part 4a: Welfare

Consumer derives

- expected utility from first-best contract equal to : $U_{FB} = \mathbb{E}_l[v(w^{FB}(l); l)]$
- expected utility from null contract equal to : $U_0 = \mathbb{E}_l[v(w_0; l)]$
- expected utility from contract x equal to : $U_x = \mathbb{E}_l[\underbrace{u(y^*(l), m^*(l); l)}_{u^*(l)}]$

Model part 4a: Welfare

Consumer derives

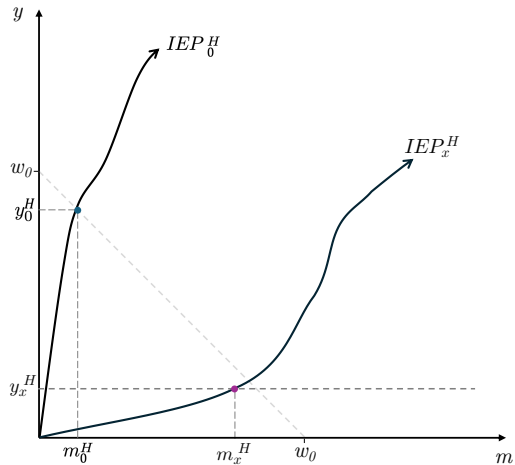
- expected utility from first-best contract equal to : $U_{FB} = \mathbb{E}_l[v(w^{FB}(l); l)]$
- expected utility from null contract equal to : $U_0 = \mathbb{E}_l[v(w_0; l)]$
- expected utility from contract x equal to : $U_x = \mathbb{E}_l[\underbrace{u(y^*(l), m^*(l); l)}_{u^*(l)}]$

- ▶ define $\tilde{w}_x(l) \equiv e(u^*(l); l)$ = minimum amount of money needed to reach utility level u^* in state l when $p_y = p_m$

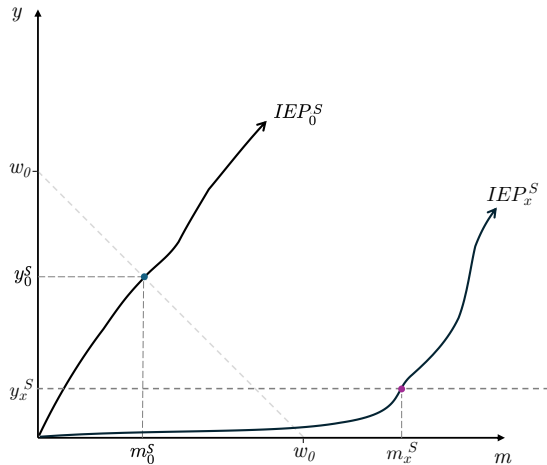
$$\Rightarrow U_x = \mathbb{E}_l[\underbrace{v(\tilde{w}_x(l); l)}_{u^*(l)}]$$

Pictures part 4a: Welfare

$l = \text{Healthy}$

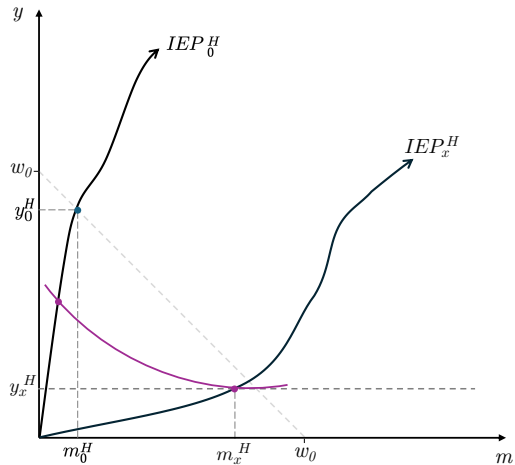


$l = \text{Sick}$

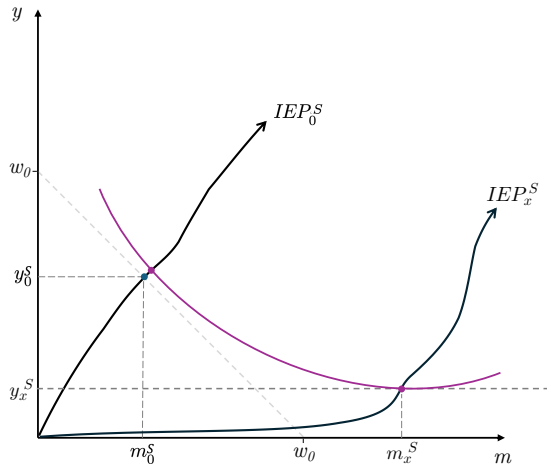


Pictures part 4a: Welfare

$l = \text{Healthy}$

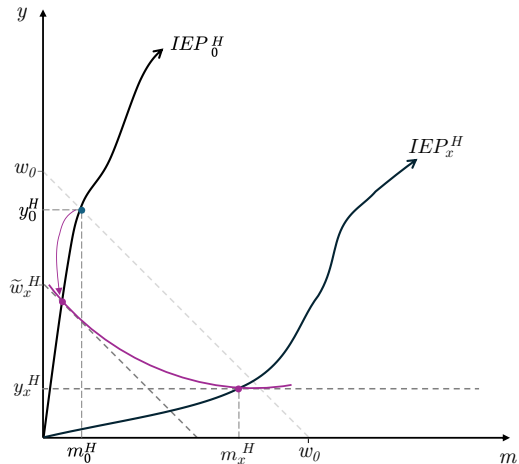


$l = \text{Sick}$

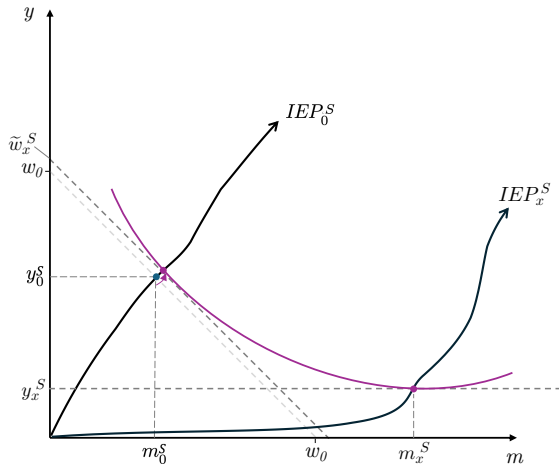


Pictures part 4a: Welfare

$l = \text{Healthy}$

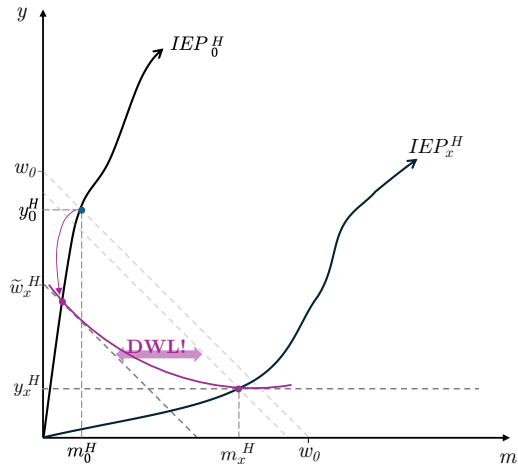


$l = \text{Sick}$

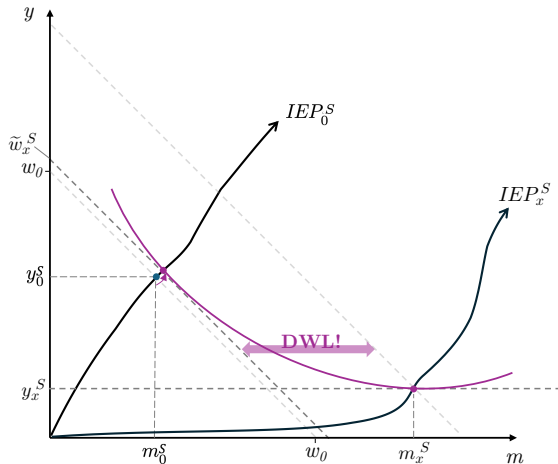


Pictures part 4a: Welfare

$l = \text{Healthy}$

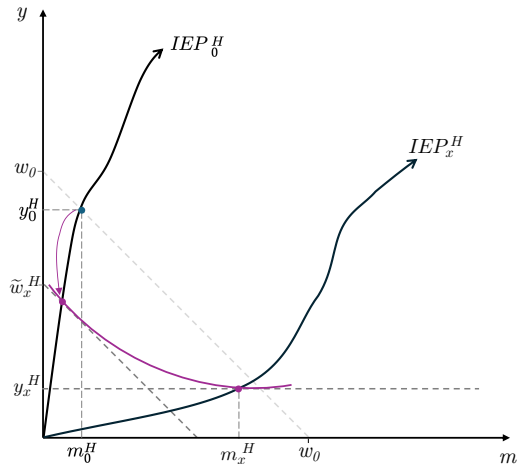


$l = \text{Sick}$

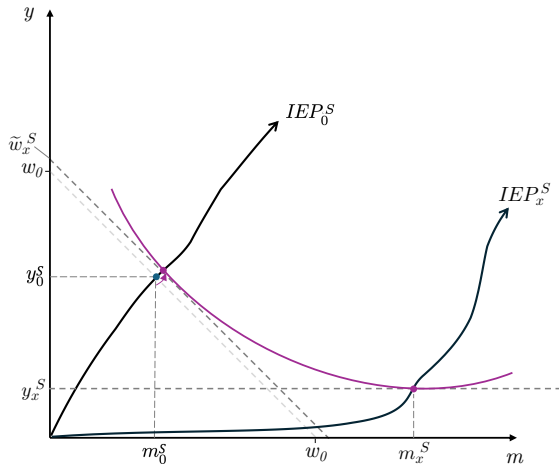


Pictures part 4a: Welfare

$l = \text{Healthy}$

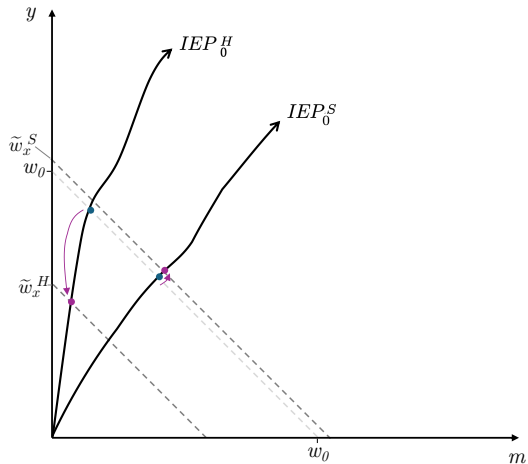


$l = \text{Sick}$



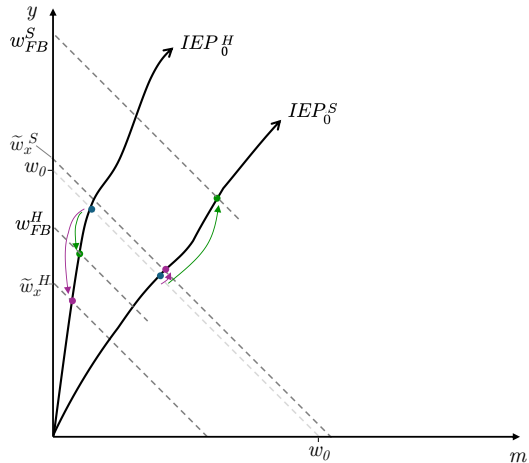
Pictures part 4a: Welfare

Ordinal preferences



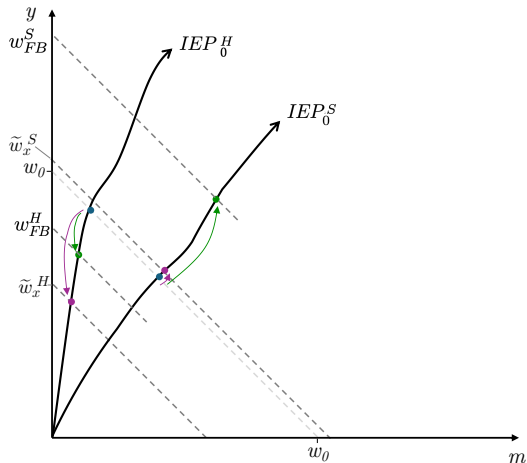
Pictures part 4a: Welfare

Ordinal preferences

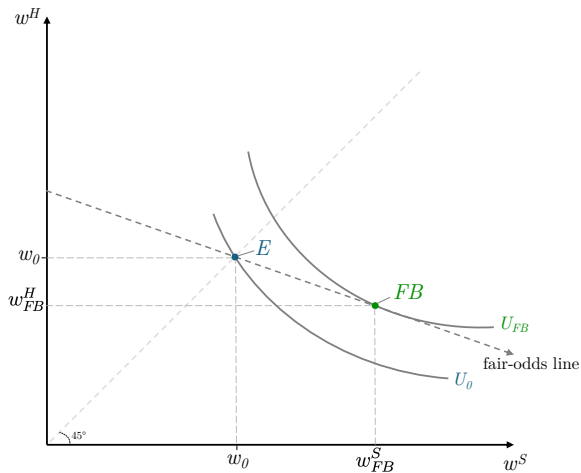


Pictures part 4a: Welfare

Ordinal preferences

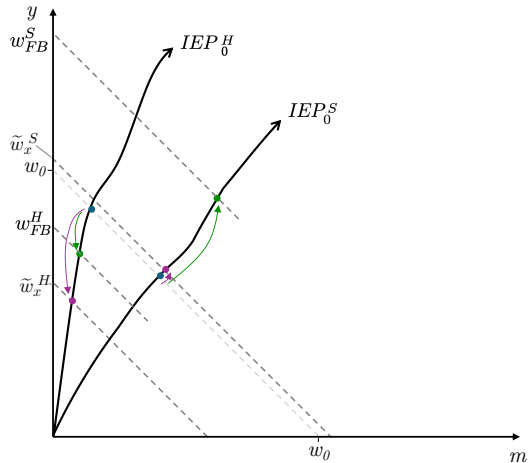


Risk preferences

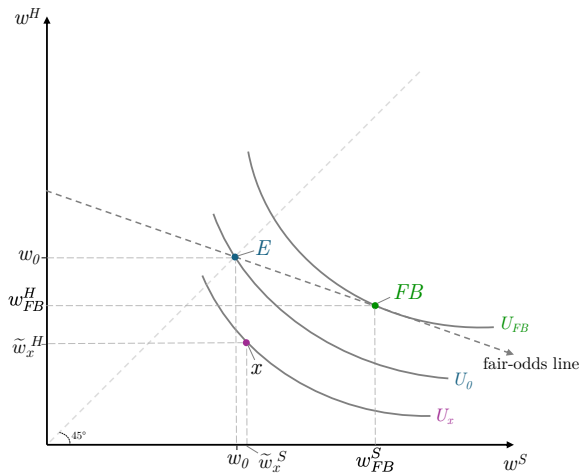


Pictures part 4a: Welfare

Ordinal preferences

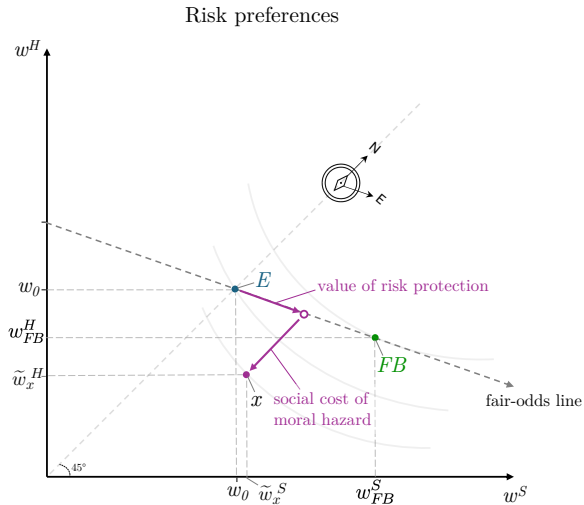


Risk preferences



Pictures part 4b: Welfare decomposition

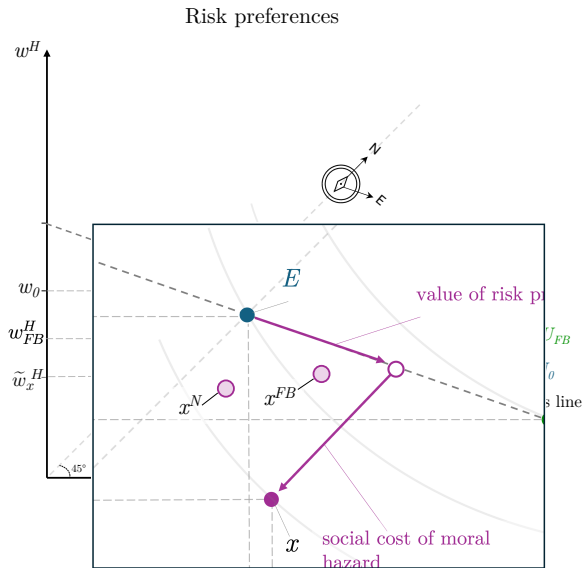
$$V(x) = \underbrace{\Psi(x)}_{\text{value of risk protection}} - \underbrace{SCMH(x)}_{\text{social cost of moral hazard}}$$



Pictures part 4b: Welfare decomposition

$$V(x) = \underbrace{\Psi(x)}_{\text{value of risk protection}} - \underbrace{SCMH(x)}_{\text{social cost of moral hazard}}$$

- x^N : hold m fixed at m_0 level
- x^{FB} : hold m fixed at m_{FB} level
- x : equilibrium m

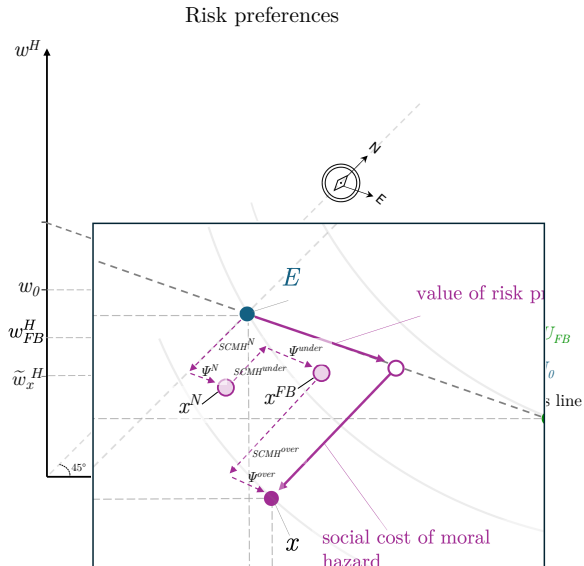


Pictures part 4b: Welfare decomposition

$$V(x) = \underbrace{\Psi(x)}_{\text{value of risk protection}} - \underbrace{SCMH(x)}_{\text{social cost of moral hazard}}$$

- x^N : hold m fixed at m_0 level
- x^{FB} : hold m fixed at m_{FB} level
- x : equilibrium m

- $E \rightarrow x$
 $\equiv$
 $E \rightarrow x^N \rightarrow x^{FB} \rightarrow x$
 - and each step can be broken
 into Ψ and $SCMH$



Empirical magnitudes

Utility specification	Decomposition of welfare (avg. fraction of total)			
	Ψ_N	Ψ_{under}	Ψ_{over}	$SCMH$
Marone & Sabety 2022	0.92	0.10	<0.01	-0.02
Einav et al. 2013 (<i>additive</i>)	0.93	0.09	<0.01	-0.02
Einav et al. 2013 (<i>multiplicative</i>)	0.95	0.03	0.04	-0.03
Ho & Lee 2023	1.12	0.01	0.03	-0.16

Notes: For each household's socially optimal contract

- Ψ_N = Welfare if consumer prevented from changing behavior in response to insurance
- $\Psi_{under} + \Psi_{over}$ = Risk protection value of moral hazard utilization

Empirical magnitudes

Utility specification	Decomposition of welfare (avg. fraction of total)			
	Ψ_N	Ψ_{under}	Ψ_{over}	$SCMH$
Marone & Sabety 2022	0.92	0.10	<0.01	-0.02
Einav et al. 2013 (<i>additive</i>)	0.93	0.09	<0.01	-0.02
Einav et al. 2013 (<i>multiplicative</i>)	0.95	0.03	0.04	-0.03
Ho & Lee 2023	1.12	0.01	0.03	-0.16

Notes: For each household's socially optimal contract

- Ψ_N = Welfare if consumer prevented from changing behavior in response to insurance
- $\Psi_{under} + \Psi_{over}$ = Risk protection value of moral hazard utilization

1. Introduction

2. Model & empirical magnitudes

3. Discussion & conclusion

Implications

⇒ In centrally planned health insurance, target the **right** level of utilization: m_{FB} not m_N

↳ key thought experiment :

[no] would consumer have been willing to pay for this absent insurance?

[yes] would consumer have been willing to pay for this under first-best income?

↳ in states with low probability and/or high marginal indirect utility of income,
→ under-utilization likely absent insurance → low (or zero) optimal cost-sharing

- Unlike with **selection**, no reason to think govt has an advantage over private firms in realm of **moral hazard** and optimal insurance design

↳ key market failure is **ex-post asymmetric info** ; govt has no magic power here

↳ planner should learn from competitive private markets if and where they exist
→ some relevant clues: critical illness insurance, hospitalization insurance

Conclusion

Consumer's **behavioral response to insurance** may be—and is likely to be—an important part of the mechanism through which second-best insurance provides risk protection

No insurance markets

Autarky

No moral hazard

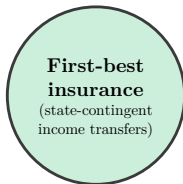


m_0

Complete markets

Competitive outcome in first-best world

No moral hazard



m_{FB}^S

Incomplete markets

Competitive outcome in second-best world

Moral hazard



m_x^S

under-utilization

over-utilization

moral hazard utilization

Slutsky income vs substitution effects

Back

$l = \text{Sick}$

